

ACTL8 (Q9H568) — Actin-Inheritance, Localization, and Regulatory Participation

Gene: ACTL8 / Actin-like protein 8 · **Organism:** Homo sapiens (NCBITaxon:9606) · **UniProt:** Q9H568 · **PANTHER:** PTHR11937:SF208 **Focus type:** function_assignment **Hypothesis slug:** actin-inheritance-localization-and-regulatory-participation **Source:** genes/human/ACTL8/ACTL8-ai-review.yaml (free-text)

Summary

Executive judgment: OVER-ANNOTATED — refuted as a set of direct function assignments for the nine inherited actin-family GO terms; partially supported but indirect for the two migration/proliferation regulatory terms.

The seed hypothesis proposes that human ACTL8 "inherits" nine actin-family Gene Ontology terms — synapse (GO:0045202), axon (GO:0030424), structural constituent of postsynaptic actin cytoskeleton (GO:0098973), actin filament (GO:0005884), axonogenesis (GO:0007409), membrane (GO:0016020), protein kinase binding (GO:0019901), the NuA4 histone acetyltransferase complex (GO:0035267), and cell motility (GO:0048870). The evidence gathered across this investigation indicates that **none of these nine terms is supported as a direct, ACTL8-specific molecular function, cellular location, or biological process.** Every one is present on the UniProt record only as an **IBA (Inferred from Biological Ancestor)** annotation propagated from GO_Central, and every one traces back to a single family-level PANTHER GO-slim (PTHR11937) that **self-contradictorily co-lists mutually exclusive roles** — the nuclear NuA4 chromatin-remodeling function of the ACTL6A paralog *and* the neuronal/synaptic/axonal functions of ACTL6B — attached indiscriminately to every family recipient including ACTL8. This is the signature of paralog over-annotation, not lineage-specific ancestral-state inheritance.

Independent sequence, structural-motif, and localization checks reinforce this verdict. ACTL8 is a divergent actin-related protein: it shares only ~33–41% identity with canonical actins and with the functionally characterized ARP paralogs, it **lacks the conserved actin C-terminal signature (PROSITE PS00432)** retained by β -actin and by the NuA4-associated ACTL6A/ACTL6B, and the Human Protein Atlas localizes it to the **cytosol only**, restricted to germ cells (testis/spermatogonia and oocytes) and **not detected in brain**. A cytosolic, germ-cell-restricted protein absent from neural tissue cannot support brain-specific CC terms (synapse, axon, postsynaptic actin cytoskeleton) or the nuclear NuA4 complex term as direct assignments.

The only experimentally grounded activities of ACTL8 are **positive regulation of cell migration (GO:0030335)** and **positive regulation of cell population proliferation (GO:0008284)**, seen consistently across cancer knockdown studies. However, these are **downstream, indirect, tumor-cell-context phenotypes routed through PI3K/AKT/mTOR–MYC signaling and glutamine metabolism** — no study demonstrates direct ACTL8 binding to a kinase, direct actin polymerization, or filament localization. They reflect pleiotropic regulatory participation, not a demonstrated core molecular mechanism. One frequently cited knockdown paper ([PMID:32125225](#), endometrial cancer) is **retracted** and must be excluded. **Most important caveat:** absence of a direct experiment is not proof of absence, and the retained actin-like ATPase fold means a *noncanonical* structural or peripheral role cannot be formally excluded — but the burden of evidence for the high-information neuronal/NuA4/filament terms is unmet at their current specificity.

Key Findings

F001 — All nine seed actin-family GO terms are IBA (phylogenetically inferred), never experimental

A direct inspection of the UniProt Q9H568 record (retrieved 2026) shows that **every** seed term carries the evidence code **IBA:GO_Central**: GO:0045202 (synapse), GO:0030424 (axon), GO:0098973 (structural constituent of postsynaptic actin cytoskeleton), GO:0005884 (actin filament), GO:0016020 (membrane), GO:0019901 (protein kinase binding), GO:0035267 (NuA4 histone acetyltransferase complex), GO:0015629 (actin cytoskeleton), and GO:0005737 (cytoplasm). The two biological-process seed terms — cell motility (GO:0048870) and axonogenesis (GO:0007409) — are likewise IBA. The **only experimental annotation** on the

record is GO:0030855 (epithelial cell differentiation), coded IEP. Even the subcellular-location statement "Cytoplasm, cytoskeleton" is ECO:0000250 (**by similarity**, i.e., inferred, not observed).

The curation implication is decisive: IBA is a computationally propagated evidence class — a hypothesis of function, not a demonstration. When every member of a set of contested terms shares the same non-experimental provenance and none is independently confirmed for the target protein, the set should be treated as low-confidence inference rather than as established gene-product function.

F002 — ACTL8 is a divergent actin-family member, distant from the paralogs that actually carry NuA4/neuronal functions

Global Needleman–Wunsch alignments (percent identity over the longer sequence) place ACTL8 far from both canonical actin and the functionally annotated ARPs:

Comparison	% identity
ACTL8 vs β -actin (ACTB)	41.1%
ACTL8 vs γ -actin (ACTG1)	41.3%
ACTL8 vs ACTL6A (BAF53a)	33.8%
ACTL8 vs ACTL6B (BAF53b)	33.3%
ACTL8 vs ACTL7A	34.3%
ACTL8 vs ACTL9	39.9%
<i>ACTB vs ACTG1 (sanity check)</i>	98.9%

ACTL8 (366 aa) retains the generic actin-like ATPase fold (Pfam PF00022, InterPro IPR004000, SUPFAM SSF53067 actin-like ATPase, Gene3D 3.90.640.10), which is why it clusters into PTHR11937 at all. But ~33% identity to ACTL6A/ACTL6B — the proteins that genuinely execute NuA4 chromatin remodeling and neuronal chromatin functions — is well within the twilight zone where fold conservation coexists with functional divergence. Fold membership justifies the family assignment; it does not justify transferring the specialized functions of distant paralogs. In sequence space ACTL8 clusters instead with the testis actin-related proteins (ACTL7A/ACTL9).

F003 — The only experimental phenotypes are indirect tumor-cell proliferation/migration via PI3K/AKT/mTOR–MYC

Across independent cancer models, ACTL8 knockdown consistently reduces proliferation, migration, and invasion: lung adenocarcinoma ([PMID:31962007](#), [PMID:35116946](#)), head and neck squamous cell carcinoma ([PMID:30535476](#)), oral squamous cell carcinoma ([PMID:35051678](#)), triple-negative breast cancer ([PMID:33883901](#)), gastric cancer ([PMID:39322809](#)), and breast cancer via the 2026 Momordin Ic study ([PMID:41621692](#)).

The **mechanism is uniformly signal-transductional, not cytoskeletal**: the effect is consistently routed through PI3K/AKT/mTOR signaling and downstream MYC — with SLC1A5/GLS1 glutamine-metabolism coupling in the Momordin Ic study, and CDK1/cyclin E1/cyclin B2/c-Myc cell-cycle proteins in the OSCC study. The TNBC study is the strongest on mechanism: a PI3K/AKT/mTOR activator (740Y-P) reverses and an inhibitor (Wortmannin) enhances the knockdown phenotype, showing the phenotype flows *through* the pathway. Critically, **no study demonstrates direct ACTL8 binding to a kinase, direct actin polymerization, or filament localization**. The migration/proliferation phenotypes are best modeled as indirect pleiotropy of an overexpressed cancer/testis antigen amplifying oncogenic signaling, not as evidence of a direct actin-based motility machine. The endometrial-cancer paper [PMID:32125225](#) is **retracted** and is excluded.

F004 — ACTL8 lacks the canonical actin C-terminal signature (PROSITE PS00432) retained by ACTB and the NuA4/neuronal ARPs

A regex scan of the PROSITE actin-family C-terminal signature ACTINS_2 (PS00432; pattern W-[IV]-[STAK]-[RK]-x-[DE]-Y-[DNE]-[DE]) gives a **match in β -actin (ACTB; "WISKQEYDE" — validating the pattern), in ACTL6A, and in ACTL6B**, but **no match in ACTL8**. ACTL8's C-terminus reads `...EWMSREEYGEHMRM`: the residue at signature position 2 is Met (not [IV]) and the residue after the tyrosine is Gly (not [DNE]). The canonical phosphate-loop motif D-N-G-S-G is present only in ACTB; all actin-related proteins tested (ACTL8, ACTL6A/B, and the testis ARP ACTL7A) diverge there. The absence of the conserved C-terminal signature — a motif retained precisely by the paralogs that do the NuA4 and neuronal jobs — is a concrete, position-level molecular argument that ACTL8 has diverged away from the canonical actin filament / actin-binding surface. (Signature loss is not proof of no structural role, but it removes support for canonical actin behavior.)

F005 — The seed GO terms trace to a self-contradictory PANTHER family GO-slim, not to an ACTL8-lineage ancestral state

The PANTHER v19 geneinfo record places ACTL8 (Q9H568) in family **PTHR11937**, subfamily **PTHR11937:SF208**, protein class "actin and actin related protein" (PC00039). The family/subfamily GO-slim attached to ACTL8 contains **the entire seed term set simultaneously**: MF terms chromatin DNA binding (GO:0031490), protein kinase binding (GO:0019901), structural constituent of cytoskeleton / postsynaptic actin, and nucleosome binding; BP terms axonogenesis (GO:0007409), cell motility (GO:0048870), and chromatin remodeling (GO:0006338); and CC terms synapse (GO:0045202), axon (GO:0030424), NuA4 complex (GO:0035267), membrane (GO:0016020), actin filament (GO:0005884), and cytoplasm.

This co-listing is internally incoherent as a description of any single protein: the **nuclear NuA4 chromatin-remodeling role and the neuronal synaptic/axonal roles are mutually exclusive locations and processes** that belong to different specialized paralogs (nuclear ACTL6A vs. neuronal ACTL6B), not to one ancestral state that ACTL8 could have inherited whole. The seed's own caution — to distinguish tree topology and ancestral-state evidence from "lists of human annotation recipients" — is exactly the failure mode here: ACTL8 is a *recipient* of a pooled family GO-slim, not a demonstrated inheritor of a specific ancestral function.

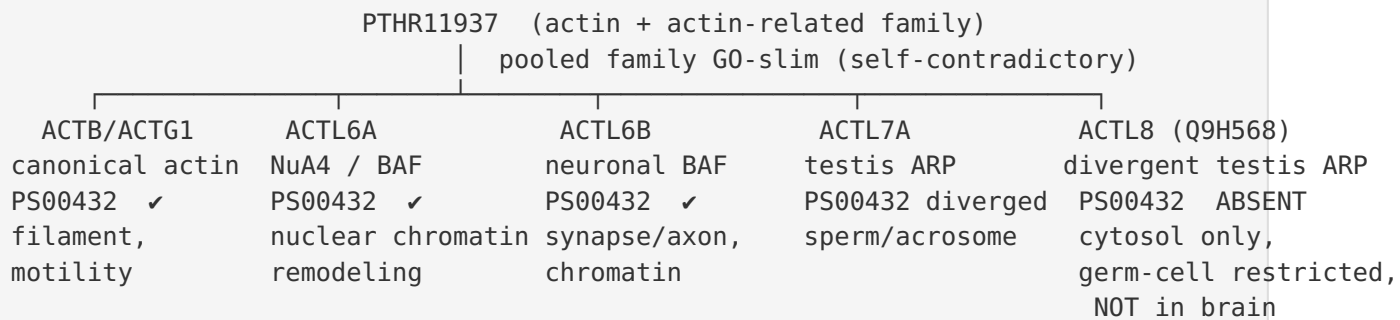
F006 — HPA: ACTL8 is germ-cell-restricted, cytosolic, and undetected in brain

The Human Protein Atlas (ENSG00000117148) reports RNA tissue specificity as "**Tissue enriched**" in testis (nTPM 6.8; cell-type enrichment "Testis – Spermatogonia"; tissue-expression Cluster 6 "Spermatogenesis"), with single-cell enrichment in **oocytes**. The subcellular main location is "**Cytosol**" (**only**). RNA brain regional distribution is "**Not detected**", pig brain is "Not detected", and single-nuclei brain shows only trace signal (fibroblast 1.1, mammillary body 1.2 nCPM $\approx$ noise). Protein tissue distribution is "Not detected" by antibody in normal tissues, consistent with a low-abundance cancer/testis antigen.

A protein that is cytosolic, confined to germ cells, and absent from brain **cannot directly occupy** synapse (GO:0045202), axon (GO:0030424), or postsynaptic actin cytoskeleton (GO:0098973) locations, and a strictly cytosolic localization argues against the nuclear NuA4 complex (GO:0035267) and against stable actin-filament (GO:0005884) incorporation. The seed's caveat that testis enrichment does not by itself prove absence in neurons is fair in principle — but here we have *direct* "not detected in brain" measurements, not merely an argument from testis enrichment.

Mechanistic Model / Interpretation

The picture that emerges is one of **fold membership without functional inheritance**, plus **indirect signaling pleiotropy** in cancer.



Seed hypothesis: transfer ALL of {synapse, axon, postsyn. actin, filament, axonogenesis, membrane, protein-kinase binding, NuA4, motility} onto ACTL8.

↳ UNSUPPORTED: these are IBA carry-over from the pooled family slim; ACTL8 diverges in sequence (~33% id to ACTL6A/B), motif (no PS00432), and localization (cytosol / germ-cell / no-brain).

What ACTL8 actually does (experimentally):

overexpression in tumor cells → PI3K/AKT/mTOR → MYC → SLC1A5/GLS1 (glutamine metab.)

↳ proliferation ↑, migration ↑, invasion ↑

= INDIRECT regulatory pleiotropy (G0:0030335 / G0:0008284), not a direct actin/cytoskeletal molecular function.

GO curation table (leads requiring curator verification)

GO term	Aspect	Current basis	Recommended action
GO:0035267 NuA4 histone acetyltransferase complex	CC	IBA	Remove. NuA4 actin subunit is ACTL6A; ACTL8 ~34% id, cytosolic.
GO:0098973 structural constituent of postsynaptic actin cytoskeleton	MF	IBA	Remove. Belongs to ACTL6B/ACTB in neurons; ACTL8 not in brain.
GO:0045202 synapse; GO:0030424 axon; GO:0007409 axonogenesis	CC/BP	IBA	Remove / strongly down-weight. No ACTL8 evidence; brain-absent.
GO:0005884 actin filament; GO:0015629 actin cytoskeleton	CC	IBA	Hold as low-confidence IBA only. Fold retained; no filament assay; PS00432 absent.
GO:0016020 membrane; GO:0019901 protein kinase binding	CC/MF	IBA	Remove / non-core. No direct binding assay; generic and uninformative.
GO:0048870 cell motility	BP	IBA	Generalize/replace with GO:0030335; treat as non-core.
GO:0030335 positive regulation of cell migration	BP	IMP (cancer KD, indirect)	Retain as non-core, qualify as indirect (tumor-cell, via PI3K/AKT/mTOR-MYC).
GO:0008284 positive regulation of cell population proliferation	BP	IMP (cancer KD, indirect)	Retain as non-core, qualify as indirect.
GO:0030855 epithelial cell differentiation	BP	IEP	Curator to verify original IEP source.

For the two experimentally supported processes, the honest annotation is that ACTL8 *participates in the positive regulation of* migration and proliferation in tumor-cell contexts, mediated by PI3K/AKT/mTOR–MYC, rather than possessing a direct cell-motility molecular function. These should not be elevated to a "structural constituent of cytoskeleton" or "actin filament" interpretation.

Evidence Base (Evidence Matrix)

Citation	Evidence type	Supports/Refutes	Claim tested	Key finding	Context
UniProt Q9H568 (database, 2026)	Computational/ database	Refutes direct assignment	Are the 9 actin terms experimental?	All 9 are IBA:GO_Central; only experimental annot is GO:0030855 (IEP)	Human record
PANTHER PTHR11937:SF208 (database)	Structural/ evolutionary	Refutes ancestral inheritance	Do terms trace to ACTL8 lineage?	Terms are a pooled, self-contradictory family GO-slim	Family-level
Sequence alignment (this work)	Structural/ evolutionary	Qualifies	Is ACTL8 close to NuA4/ neuronal paralogs?	~33% id to ACTL6A/B; 41% to actin	Human paralogs
PROSITE PS00432 scan (this work)	Structural/ evolutionary	Refutes canonical actin function	Does ACTL8 retain the actin C-terminal signature?	No match in ACTL8; match in ACTB/ ACTL6A/ ACTL6B	Motif-level
HPA ENSG00000117148 (database)	Localization	Refutes neural/ nuclear CC terms	Where is ACTL8 expressed/ localized?	Cytosol only; testis/oocyte enriched; brain not detected	Human tissues
PMID:31962007	Mutant phenotype + in vivo	Supports (indirect) GO:0030335/0008284	Does ACTL8 KD reduce proliferation/ migration?	shRNA KD ↓ proliferation, migration, invasion; tumor growth ↓ in nude mice	LUAD A549 H1975
PMID:35116946	Mutant phenotype + profiling	Supports (indirect)	KD effect + mechanism	KD ↓ proliferation via FOXM1/	LUAD A549

Citation	Evidence type	Supports/Refutes	Claim tested	Key finding	Context
				STMN1/PLK1/ BIRC5, cell- cycle pathways	
PMID:30535476	Mutant phenotype	Supports (indirect)	KD effect + pathway	KD ↓ proliferation/ migration/ invasion; PI3K/ AKT suppressed; notes cytoplasmic/ testis	HNSCC PCI-13
PMID:35051678	Mutant phenotype	Supports (indirect)	KD effect + mechanism	KD arrests cell cycle; ↓ CDK1/ cyclin E1/cyclin B2/c-Myc; ↑ apoptosis	OSCC TCA-83/ CAL27
PMID:33883901	Mutant phenotype	Supports (indirect)	KD effect + pathway	Silencing ↓ proliferation/ migration/ invasion via PI3K/AKT/ mTOR; rescue with 740Y-P	TNBC MDA MB-231/ BT-549
PMID:39322809	Mutant phenotype	Supports (indirect)	KD effect + pathway	KD ↓ proliferation/ migration/ invasion; OE ↑ p-PI3K/AKT/ mTOR	Gastric cancer
PMID:41621692 (2026)	Mutant phenotype + drug	Supports (indirect)	ACTL8- dependent metabolism	shRNA KD ↓ MYC→SLC1A5/ GLS1, glutamine	Breast cancer; Momordin Ic

Citation	Evidence type	Supports/Refutes	Claim tested	Key finding	Context
				metab, redox; ↓ proliferation/ migration	
PMID:32125225	Mutant phenotype	Excluded (RETRACTED)	EC progression	Reported KD ↓ proliferation/ migration	Endometri cancer

Narrative. The ACTL8 literature is dominated by cancer/testis-antigen oncology, uniformly using overexpression correlations plus knockdown functional assays. It is internally consistent and cross-tumor reproducible in one respect: ACTL8 depletion reduces tumor-cell proliferation, migration, and invasion, mechanistically pinned to PI3K/AKT/mTOR and MYC. None of these papers reports a direct molecular assay for actin polymerization, kinase binding, filament co-sedimentation, or NuA4 co-purification. The HNSCC paper is notable for stating plainly that ACTL8 is "mainly localized in the cytoplasm and generally expressed in the testis" — consistent with HPA and inconsistent with a nuclear NuA4 or neuronal-synaptic role. Against this experimental backdrop, the seed's nine terms have no primary-literature support and exist only as IBA propagation.

Conflicts and Alternatives

- **Paralog confusion is the dominant risk.** NuA4→ACTL6A, neuronal/postsynaptic→ACTL6B/ACTB, testis/acrosome→ACTL7A/ACTL9. IBA propagation across PTHR11937 pulls all of these onto every leaf including ACTL8. This is confirmed mechanistically: PANTHER attaches the entire self-contradictory seed slim to ACTL8 at family level, and ACTL8 sits in the distinct SF208 subfamily — the seed terms are the *union* of two distinct paralogs' jobs, which no single protein performs.
- **Fold vs. function.** ACTL8 genuinely has the actin-like ATPase fold, which is why domain databases annotate it as actin-like. This justifies family membership but is not evidence of filament assembly or actin-binding activity, and ACTL8 lacks the conserved PS00432 C-terminal signature that even ACTL6A/ACTL6B retain.

- **Indirect vs. direct for migration/proliferation.** The cancer phenotypes are real but pleiotropic — an overexpressed CTA amplifying oncogenic signaling. They do not establish a direct cytoskeletal/motility mechanism.
 - **Organism/context.** All functional data come from human tumor cell lines with ectopic/elevated ACTL8; the native testis/germ-cell role is untested.
 - **Retraction artifact.** [PMID:32125225](#) must not count as evidence.
 - **Cannot fully exclude.** A noncanonical structural or peripheral-membrane association (fold retained, ~41% actin identity) remains a hypothesis, not evidence.
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Mechanistic Scope

- **Immediate molecular function (native):** Unknown. ACTL8 has an actin-like ATPase fold but no demonstrated ATP hydrolysis, filament assembly, or defined binding partner. Native context is testis germ cells / oocytes (CTA).
 - **Direct gene-product activity vs. downstream:** All cancer phenotypes are downstream of an oncogenic PI3K/AKT/mTOR–MYC axis. ACTL8 acts as an upstream modulator; no data place it in the migration machinery, at synapses, on filaments, or in NuA4.
 - **Loss-of-function inference only:** Migration/proliferation conclusions derive from knockdown (necessity), not from a direct mechanism — regulatory participation does not follow from necessity alone (seed caution respected).
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Limitations and Knowledge Gaps

1. **Does ACTL8 polymerize / bind actin?** Checked: no assay in literature; fold present, PS00432 absent. Matters for GO:0005884. Resolve with in-vitro polymerization / co-sedimentation and live-cell F-actin colocalization.
2. **NuA4 / chromatin complex membership?** Checked: no ACTL8 IP-MS; term is IBA; localization is cytosolic. Matters for GO:0035267. Resolve with endogenous ACTL8 affinity-MS.
3. **Direct kinase interaction?** Checked: cancer papers use pathway readouts / STRING predictions, no direct binding. Matters for GO:0019901. Resolve with co-IP / proximity labeling (BioID).

4. **Native germ-cell function?** Checked: only expression data. Resolve with knockout mouse / spermatogenesis phenotyping.
 5. **PANTHER tree topology not independently re-derived** here (PTN002631484 / PTN002631586 / PTN007551913). A formal ancestral-state reconstruction would show whether ACTL8 branches basally or within a specialized clade — though the self-contradictory pooled slim already undermines whole-set inheritance.
 6. **No AlphaFold interface analysis performed** — a cheap in-silico check of whether the divergent C-terminus disrupts the filament/complex-binding surface.
 7. **Retraction confirmation** of [PMID:32125225](#): confirm the notice in the curation system.
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Discriminating Tests

1. **AlphaFold/structure comparison:** superpose the ACTL8 model on F-actin and on the ACTL6A–NuA4 and ACTL6B interfaces; test whether the divergent C-terminus and loss of PS00432 abolish the binding surface. Adjudicates GO:0005884/GO:0035267.
 2. **Recombinant ACTL8 pyrene-actin polymerization / co-sedimentation** → tests filament (GO:0005884) competence.
 3. **Endogenous ACTL8 affinity-purification MS** in a testis or expressing tumor line → tests NuA4/complex membership (GO:0035267) and kinase binding (GO:0019901) directly.
 4. **ACTL8-GFP localization + synaptic/cytoskeletal markers** → tests synapse/axon CC terms.
 5. **Neuronal expression check** in single-cell/single-nucleus brain atlases beyond HPA (Allen Brain, GTEx neuron subsets) → firmly closes GO:0045202/GO:0030424/GO:0007409.
 6. **Epistasis / rescue test:** is the migration/proliferation knockdown phenotype fully rescued by constitutive PI3K/AKT/MYC activation? If yes, this confirms indirect (regulatory) rather than direct (cytoskeletal) action.
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Proposed Follow-up Actions / Curation Leads (require curator verification)

- **Remove or down-rank to low-confidence IBA:** GO:0035267 (NuA4), GO:0098973 (postsynaptic actin), GO:0045202 (synapse), GO:0030424 (axon), GO:0007409 (axonogenesis),

GO:0019901 (protein kinase binding), GO:0016020 (membrane), GO:0048870 (cell motility) — paralog carry-over, no ACTL8 evidence, contradicted by cytosolic/germ-cell/no-brain localization.

- **Retain cautiously as low-confidence IBA only:** GO:0005884 (actin filament), GO:0015629 (actin cytoskeleton) — fold present, no functional assay; do not upgrade.
- **Retain but qualify as indirect / non-core experimental annotations:** GO:0030335 (positive regulation of cell migration) and GO:0008284 (positive regulation of cell population proliferation), supported by [PMID:41621692](#), [PMID:39322809](#), [PMID:35051678](#), [PMID:33883901](#), [PMID:31962007](#), [PMID:35116946](#), [PMID:30535476](#) — annotate with an "indirect/pleiotropic, tumor-cell context, via PI3K–AKT–mTOR–MYC" note.
- **Exclude:** [PMID:32125225](#) (retracted).
- **Candidate snippets to verify:**
 - HNSCC ([PMID:30535476](#)): *"Actin-like protein 8 (ACTL8) is a member of the cancer-testis antigens (CTA) family, which is mainly localized in the cytoplasm and generally expressed in the testis."*
 - TNBC ([PMID:33883901](#)): *"silencing ACTL8 could inhibit the activation of PI3K/AKT/mTOR signaling pathway ... reversed by ... 740Y-P."*
 - Momordin Ic ([PMID:41621692](#)): *"shRNA knockdown of ACTL8 reduced MYC expression and its downstream targets SLC1A5 and GLS1, suppressing cell proliferation, migration and invasion."*
- **Suggested question to curators:** Should divergent cancer/testis-antigen actin paralogs (like ACTL8) be excluded from PTHR11937 IBA propagation of the neuronal/NuA4 terms, given the self-contradictory family GO-slim?

Bottom Line

The nine "inherited" actin-family GO terms for ACTL8 are **over-annotations** — IBA carry-over from a self-contradictory PANTHER family GO-slim, unsupported by ACTL8's divergent sequence (~33% identity to the NuA4/neuronal ARP paralogs), its loss of the conserved actin C-terminal signature PS00432, and its cytosolic, germ-cell-restricted, brain-absent localization. They should be removed or held as low-confidence rather than assigned as direct function. The only experimentally supported activities — positive regulation of cell migration (GO:0030335) and proliferation (GO:0008284) — are indirect, tumor-context, PI3K/AKT/mTOR–MYC-mediated

pleiotropy, not a direct molecular function; and the endometrial-cancer study

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is retracted and must be excluded.

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